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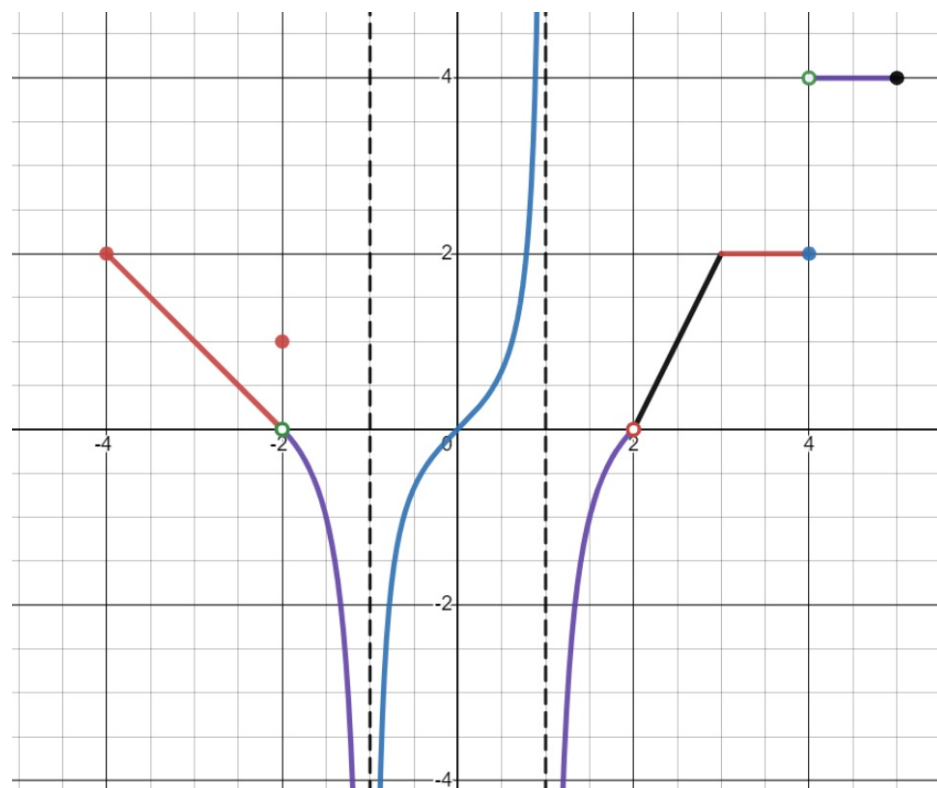
1 Functions and Limits

1.1 Math 140 Review

1. If $f(x) = \sqrt{x+5} - \sin\left(\frac{\pi}{2}x\right)$, then find the exact value of $f\left(\frac{2}{3}\right)$
2. Find the domain of each function:
 - (a) $f(x) = \sqrt{2-5x}$
 - (b) $f(x) = \frac{1}{x\sqrt{x+3}}$
 - (c) $f(x) = \sqrt[3]{2-7x}$
3. Find the difference quotient of each function:
 - (a) $f(x) = 2x^2 + x$
 - (b) $f(x) = \frac{1}{x+1}$
4. Decompose the following functions $F(x)$ into three simpler functions f, g, h so that $F(x) = (f \circ g \circ h)(x)$. Do not use the identity/trivial function for f, g or h .
 - (a) $F(x) = \sqrt{x^2 + 4}$
 - (b) $F(x) = \sin^2(2x + 1)$
 - (c) $F(x) = \tan(\cos(x^3))$

1.2 The Limit of a Function

1. The graph of $f(x)$ is drawn below. Find the following values and limits if they exist. If the limit is $+\infty$ or $-\infty$ specify which instead of “does not exist.”



(a) $\lim_{x \rightarrow -2^-} f(x)$

(i) $\lim_{x \rightarrow 1^+} f(x)$

(b) $\lim_{x \rightarrow -2^+} f(x)$

(j) $\lim_{x \rightarrow 2^-} f(x)$

(c) $\lim_{x \rightarrow -2} f(x)$

(k) $\lim_{x \rightarrow 2^+} f(x)$

(d) $f(-2)$

(l) $\lim_{x \rightarrow 2} f(x)$

(e) $\lim_{x \rightarrow -1^-} f(x)$

(m) $f(2)$

(f) $\lim_{x \rightarrow -1^+} f(x)$

(n) $\lim_{x \rightarrow 4^-} f(x)$

(g) $\lim_{x \rightarrow -1} f(x)$

(o) $\lim_{x \rightarrow 4^+} f(x)$

(h) $\lim_{x \rightarrow 1^-} f(x)$

(p) $f(4)$

1.3 Calculating Limits using Limit Laws

1. Using the rules for limits, compute the following limits. If the limit is $\pm\infty$ state which. If the limit does not exist, write DNE. *Don't forget to keep the limit notation until you take the limit!*

(a) $\lim_{x \rightarrow 3} \frac{x^2 - 4x + 3}{x^2 - 9}$

(b) $\lim_{x \rightarrow 2} \frac{x^2 + 3x - 2}{x - 1}$.

(c) $\lim_{x \rightarrow 2^+} \frac{x^2 - 2x - 8}{x^2 - 4x + 4}$

(d) $\lim_{x \rightarrow 0} \frac{x^2 - x}{x(x + 2)}$

1.4 Precise Definition of a Limit**1.5 Continuity****2 Derivatives****2.1 Derivatives and Rate of Change****2.2 Derivative as a Function****2.3 Differentiation Formulas****2.4 Derivatives of Trigonometric Functions****2.5 The Chain Rule****2.6 Implicit Differentiation****2.7 Rates of Change in Sciences****2.8 Related Rates****2.9 Linear Approximation and Differentials****3 Applications of Derivatives****3.1 Maximum and Minimum Values****3.2 The Mean Value Theorem****3.3 How Derivatives Affect the Shape of a Graph****3.4 Limits at Infinity****3.5 Summary of Curve Sketching****3.6 Graphing with Calculus****3.7 Optimization Problems****3.8 Newtons Method****3.9 Antiderivatives****4 Integrals****4.1 Areas and Distances****4.2 The Definite Integral****4.3 The Fundamental Theorem of Calculus****4.4 Indefinite Integrals and Net Change Theorem****4.5 The Substitution Rule****5 Applications of Integration****5.1 Areas Between Curves****5.2 Volumes**